INCH-POUND
MIL-PRF-49471/2A(CR)
30 November 2000
Superseding
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2 June 1995

PERFORMANCE SPECIFICATION SHEET

BATTERY, NON-RECHARGEABLE, HIGH PERFORMANCE, BA-X598/U

This specification is approved for use within the Communications Electronics Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product herein shall consist of this specification and MIL-PRF-49471(CR).

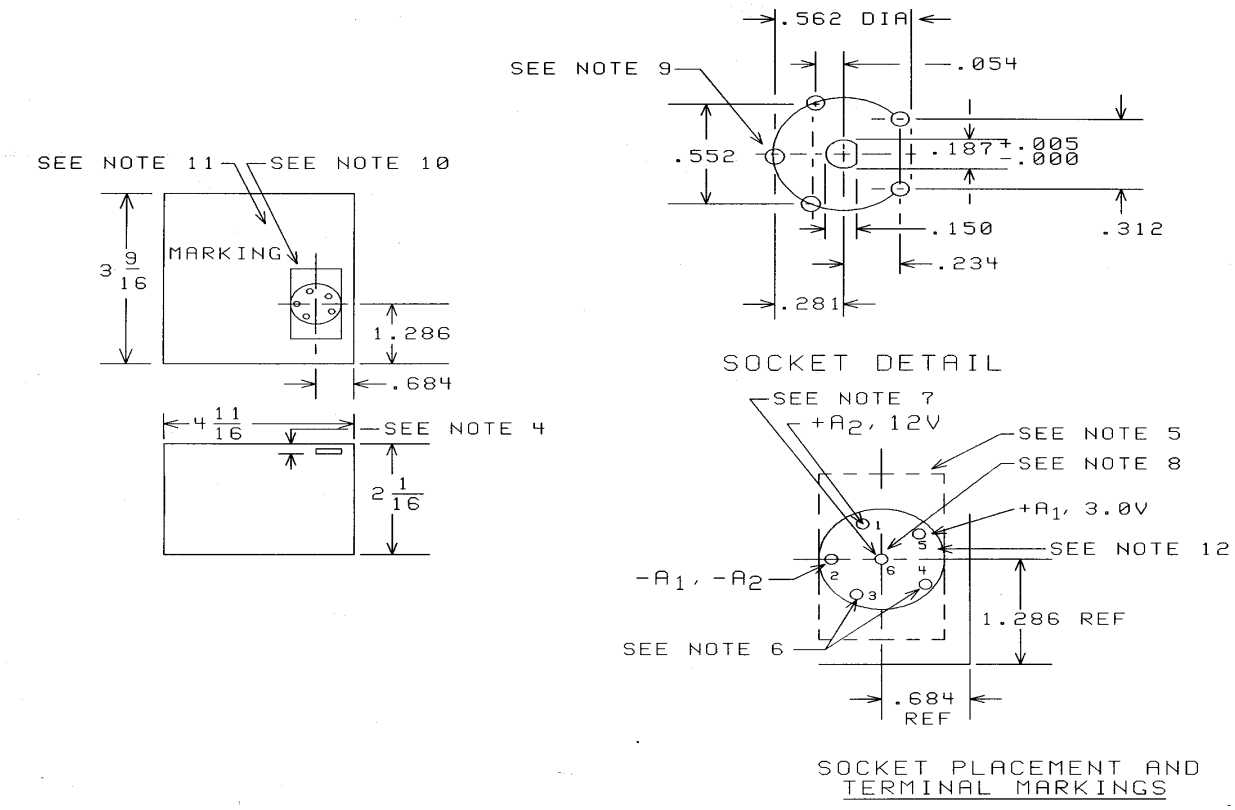

Figure 1. Battery dimensions and terminal markings.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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NOTES:

1. All dimensions are in inches.

2. Tolerances, unless otherwise specified, are ±0.005 inch for decimals.

3. Tolerances for length, width, and height of battery are ±1/16 inch for fractions.

4. The minimum depth of socket well shall be 1/2 inch.

5. The size of the insulating plate is optional.

6. Socket insert contact not required.

7. Pin-circle center.

8. "D" hole for plug positioning.

9. Five holes to fit 1/8 inch minimum pins, 7/16 inch minimum in length.

10. Socket must float 3/32 + 1/32 -1/16 inch in all directions from the specified pin circle center location in the plane of the socket from reference dimension.

11. The socket shall be supported and mounted so that the top surface of the socket shall not protrude above the adjacent outside surface of the jacket and shall not be more than 1/16 inch below adjacent outside surface of the jacket when used initially, during and after subsequent insertions of the mating plug during contractual testing.

12. Hole numbers are for reference only and need not be marked on the terminals.

REQUIREMENTS:

Dimensions and configuration: See Figure 1.

Voltage Requirements:	A_1 Unit	A_2 Unit
Maximum voltage:	3.30	16.50
Cut-off voltage:	2.00	10.00

Terminals: Five (5) hole socket type with no obstruction of any holes.

Weight (maximum): 2.25 pounds (1020 grams).

Charge protection device: Applicable.

Battery Short Circuit Test: Applicable.

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State of Charge Indicator: Not applicable.

High temperature protection: Applicable.

Complete discharge device: Applicable

Capacity tests: When the battery is tested in accordance with the methods of examination and tests of this specification, the minimum capacity test requirements shall be as specified below:

Capacity Test	Battery Capacity Requirements in Hours to 10.00 Volts
I	3.8
L	2.0
H	6.5
HT	6.3
LT	1.9
IT	3.25
LR1	52.0

Initial voltage delay: When the battery is subjected to the I, H, HT, and IT capacity tests, the time required at the beginning of discharge of the battery to reach a voltage of 10.00 volts after the specified load is applied shall not be more than 5.0 seconds. For the L and LT capacity tests the time to reach a voltage of 10.00 volts after the load is applied shall not be more than 60 seconds.

METHODS OF EXAMINATION AND TESTS:

Capacity tests:

(1) Storage:

Details on storage conditions for capacity tests are specified in basic specification.

(2) Discharge:

Cells (forced discharge): Two cells shall be discharged at 2.0 amperes at 70$\pm$5^{o}F to 2.0 volts. Each discharged cell shall then be connected in series to four fresh cells. Each string shall be force discharged at 2.0 amperes at 70$\pm$5^{o}F for 3.8 hours.

Batteries: The A_2 unit (-A_1, -A_2, to + A_2) shall be discharged as follows: For all capacity tests, except the LR1, H and HT, the batteries shall be discharged at 2.0 amperes constant current to cut-off volts (zero volts for the I test). For the H and HT capacity tests the batteries shall be discharged at 2.0 amperes constant current

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for five minutes followed by open circuit for five minutes. This cycle shall be repeated until the closed-circuit voltage reaches cut-off volts. For the LR1 test, batteries shall be discharged at 160 milliamperes to 10.5 volts, then at 2.75 watts to 1.5 volts.

Forced Discharge: After discharge to zero volts on the I capacity test, the batteries shall be force discharged at 2.0 amperes for a minimum of10 minutes.

Overload test: Batteries shall be discharged at 4 amperes.

Abuse test predischarge: Designated samples shall be discharged at 2.0 amperes for 1.9 hours.

Abuse test, pulse discharge: Batteries shall be discharged at 2.5 amperes for 1 minute followed by 1.8 amperes for 4 minutes, cycled continuously to cut-off voltage.

Closed-circuit voltage: Closed-circuit voltage measurements shall be made with a direct current voltmeter of proper range and sensitivity (see basic specification), using load resistance as shown in the following table:

UNIT RESISTANCE	MINIMUM PERMISSIBLE VOLTAGE
A_1 6.5 ohms, $\pm$ 1%	1.9 volts
A_2 5.0 ohms, $\pm$ 1%	10.0 volts (this applies to cell strings as well)

Voltage shall be above "MINIMUM PERMISSIBLE VOLTAGE" within ten (10) seconds after load is applied.

Cell closed circuit voltage test: Load resistance of 1 ohm shall be used. Voltage shall be above 2.0 volts in 5 seconds.

Socket strength test (First Article only): Before conducting the complete discharge test, the five batteries shall be subjected to the following:

Using a 0.250 $\pm$0.050 inch diameter rod, apply a force to the center of the socket and deflect it downward a distance of 0.250 +0.0625, -0 inch below the top surface of the battery.

Battery Label (only for this battery):

"DO NOT CHARGE, SHORT CIRCUIT, INCINERATE, MUTILATE OR TEST WITH U-410 ADAPTER OTHERWISE BATTERY MAY VENT OR RUPTURE RELEASING TOXIC MATERIALS"

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Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodian:
Army-CR

Preparing Activity:
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